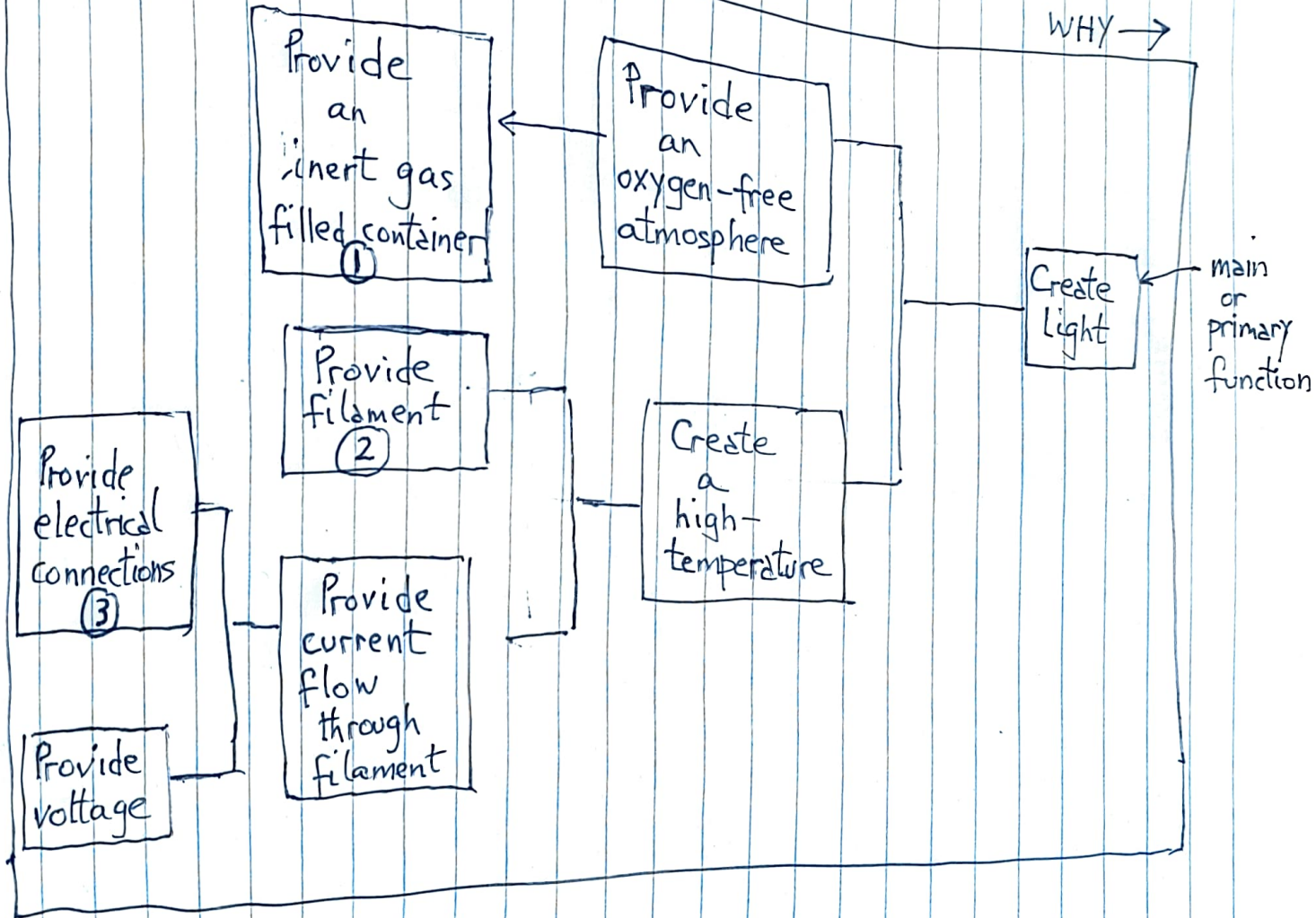




← HOW?

WHY →



Reference : See the FAST handout on  
Canvas

# Laptop

Understand how the laptop works  $\Rightarrow$  make notes

Make a list of the key sub-systems

- Microprocessor (CPU)
- Operating System
- Keyboard
- Trackpad (Mouse)
- Screen
- Storage
- Battery
- ...
- ...

Make a list of the key sub-functions

- Provide mobility
- Process information
  - select task
  - enter or input information
  - manipulate information
  - store information
  - display or output information
  - ...
- ...

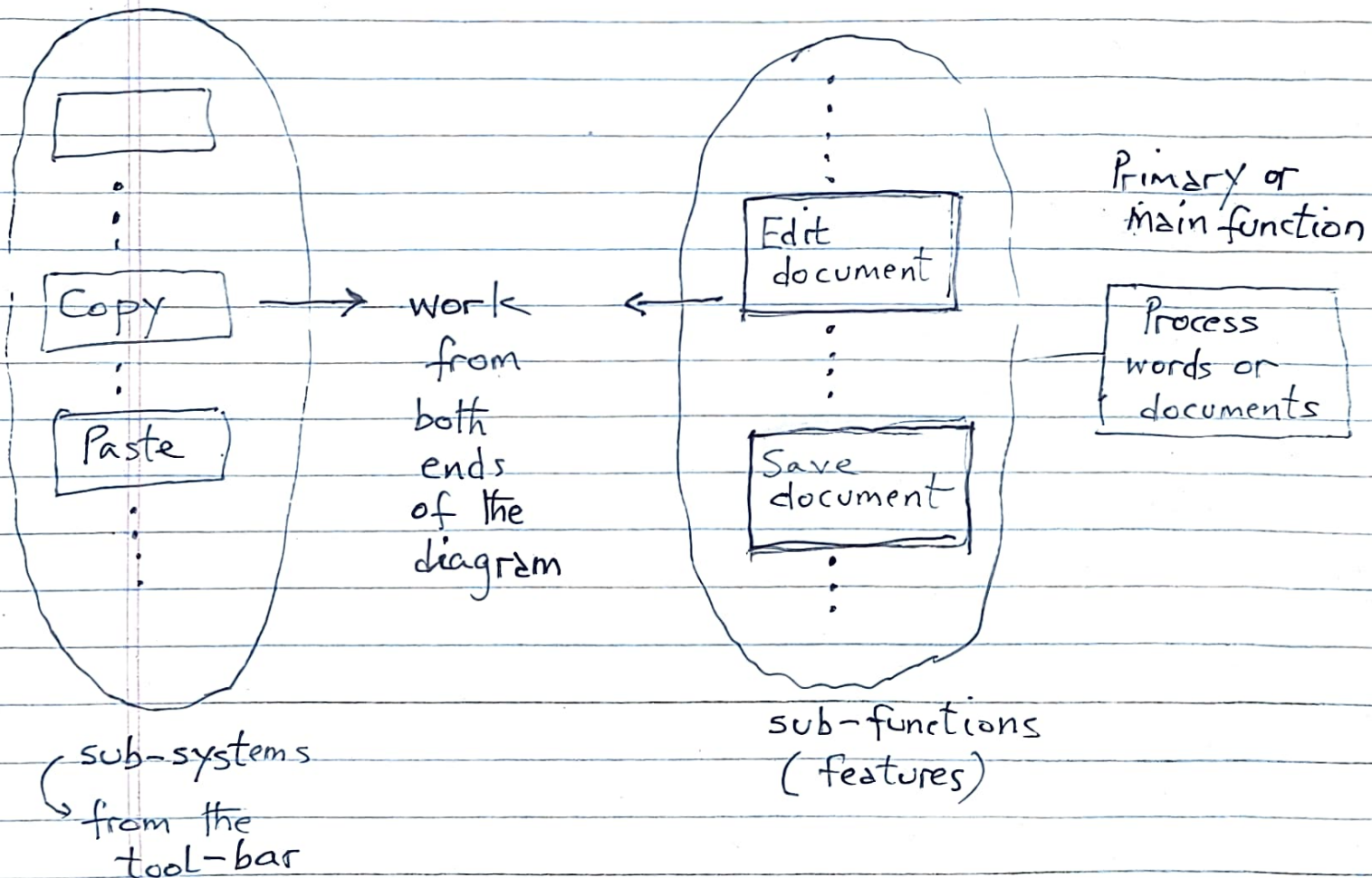
- Create a FAST as in the previous example

# Microsoft Word dissection using FAST ( $\approx 45\text{min} - 1\text{hr}$ )

Suggestion : Dissect MS Word while actually using MS Word

Use the process developed in L#3

$\left. \begin{array}{c} \vdots \\ \vdots \\ \vdots \end{array} \right\}$  steps in the process





# TIM 80C, Lecture # 4 (4/9/2026)

## Agenda:

- 0 : Complete discussion of HW # 1  
(done; see above)
- 1 : Concluding remarks on product dissection
- 2 : Function Structure (FS)
- 3 : Creating new products (and Services)
  - Function Structure (FS)
  - Morphological Matrix (MM)
  - Rank and select alternative concepts
- 4 : HW # 2 is posted & due on 4/23/2026
5. HW # 1 is now due on Friday 4/10/2026

# Function Structure (FS)

## Terminology :

"Solution neutral" representation : abstract representation of a product, which does not suggest or imply a particular solution or realization of the product.

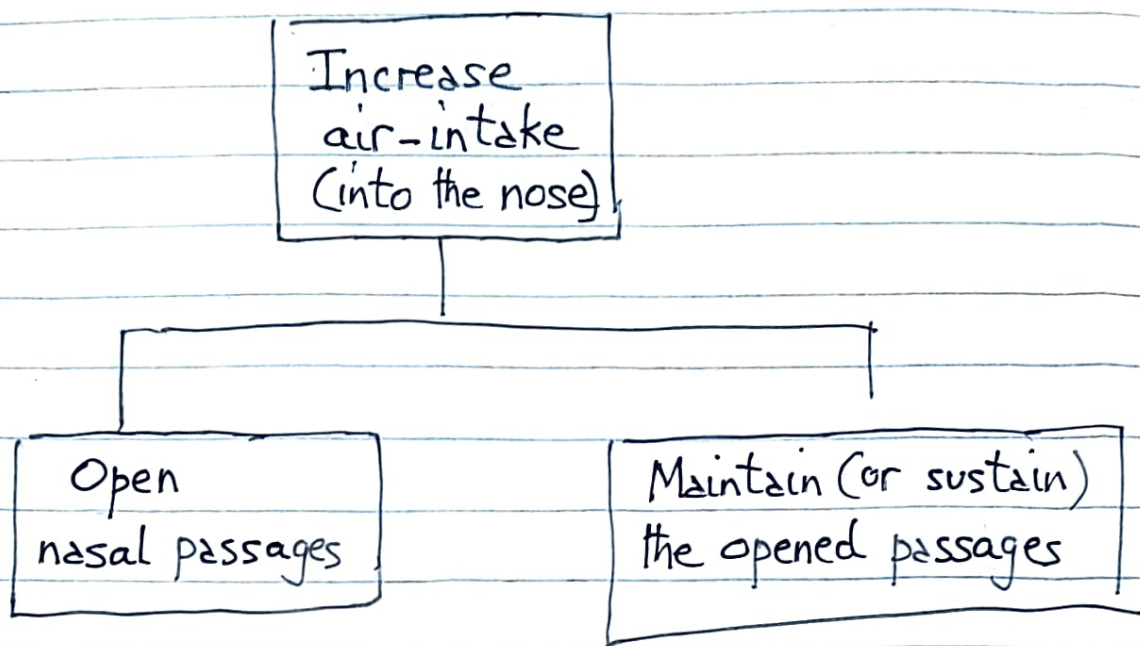
Question : Can we create a "solution neutral" representation (or mental model) of existing or new products that will enable us to generate (or create) several alternative concepts (or realizations) of the product. ?

Answer : Yes, the representation is the Function Structure (FS)



## Example : "Breathe Right" Nasal Strip

FS for the Nasal Opener :



Why is the FS }  $\Rightarrow$  enables the generation  
useful ? } of several alternative  
design concepts (or  
realizations)

→ Study The "Function Structure" handout for more examples of the FS

→ In particular, carefully study the "Automatic Tea Maker" example.



### 3. Creating New Products (aka New Product Design).

SPSP Step 1 : Define the problem : Create (or generate)

7-10 new alternative product design concepts  
as concrete realizations for a given

high-level societal or customer needs (e.g.,  
more sustainable energy sources, "green" automobiles,  
....)

Example : Nasal opener ( need : increase  
air flow into the  
nose in order to  
breathe more easily  
& sleep better, .... )

SPSP Step 2 : Plan or Approach :

step 0 : Clearly understand the customer  
needs

Create a safe, drug-free product for  
increasing the air-intake into the nose

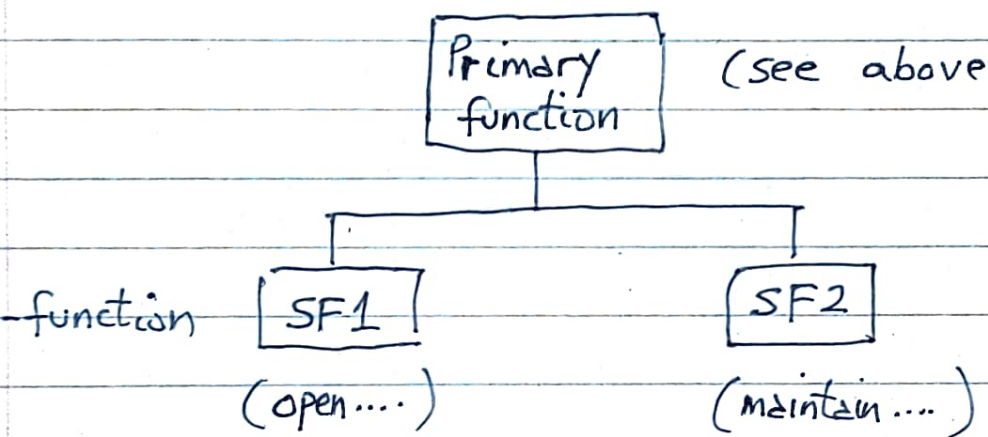


Step 1 : Clearly understand / define the primary function ( aka design intent ) of the product.

Primary function of the nasal opener:  
"Increase air-intake into the nose"

Step 2 : Create the function structure (FS) for the intended product.

The FS for the "nasal opener" was created above :



Step 3 : Create a morphological matrix (MM) showing alternative solution principles (SPs) for realizing each sub-function (SF) of the product

# Morphological Matrix for the Nasal Opener:

| solution-<br>sub-principles<br>functions →<br>↓ | Alternative solution principles |   |   |                    |                          |               |
|-------------------------------------------------|---------------------------------|---|---|--------------------|--------------------------|---------------|
| SF1:<br>open the<br>nasal<br>passages           | from the "inside"               |   |   | from the "outside" |                          |               |
| SF2:<br>maintain<br>the<br>opening              | plastic<br>tubes                | . | . | .                  | plastic<br>spring        | nose<br>-pads |
|                                                 | clip<br>①                       | . | . |                    | adhesive<br>layer<br>③ ② | . . . .       |

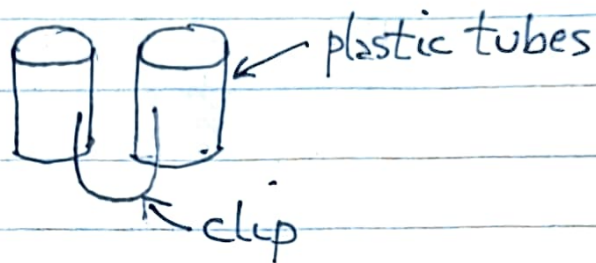
Step 4 : Generate alternative concepts by appropriately combining the solution principles for the different sub-functions

See above ↑

We generated 3 concepts shown inside the matrix

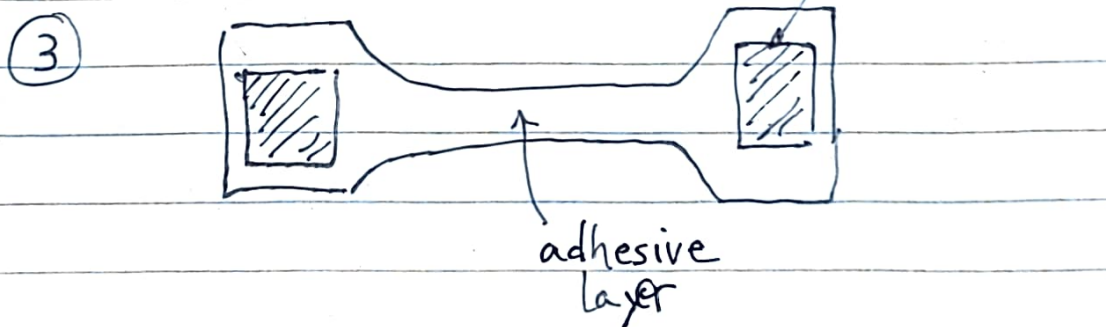
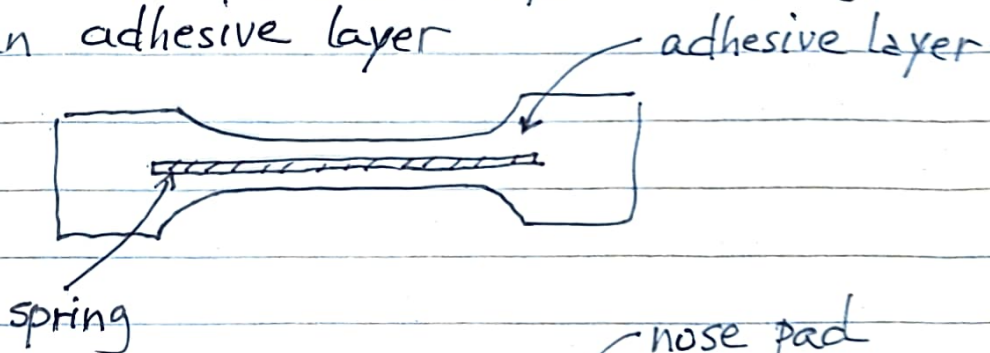
Describe the alternative concepts.

① Concept alternative 1



Two plastic tubes, one for each nostril, held together by a clip.

② existing "Breathe-Right" nasal strip:  
combination of a plastic spring &  
an adhesive layer



nose pads, one for each nostril, & an  
adhesive layer



Step 3 : Create a set of selection criteria  
in order to compare the various  
alternatives

Scale 1-5 (5=best)

| Assessment<br>criteria         | Concept<br>① | Concept<br>② | Concept<br>③ |
|--------------------------------|--------------|--------------|--------------|
| Performance<br>(effectiveness) | 4-5          | 4-5          | 2-3          |
| Price<br>attractiveness        | 1-2          | 1-2          | 4-5          |
| Safety                         | 1-2          | 5            | 5            |